Hamiltonian Systems and Celestial Mechanics

Program

June 8-12, 2026

1 List of Talks

Monday

- *Scattering in the positive energy three-body problem* — Richard Moeckel
- *When the space of quadratic forms meets Kepler's problem* — Jaime Burgos-García
- *The natural decomposition of the Jacobi equation* — Ezequiel Maderna
- *Degeneracy of four-body central configurations* — Zhifu Xie
- *Bifurcations of Centered Symmetric Central Configurations* — Michelle Gonzaga dos Santos
- *Conformal Relative Equilibria for Scaling Symmetries and Central Configurations* — Manuele Santoprete

Tuesday

- *Symmetry and N-body dynamics deformation from the plane to the sphere* — Cristina Stoica
- *Collinear motions of three bodies in a three-sphere* — Connor Jackman
- *On relative equilibria in the three-body problem on the two-dimensional hyperbolic space* — Shuqiang Zhu
- *The Veronese Geometry of Dziobek Configurations and Generic Finiteness for Homogeneous Potentials* — Thiago Dias Oliveira Silva
- *Superintegrability and choreographic solutions in quadratic dihedral n-body Hamiltonians* — Adrián Escobar
- *Homogeneous Potentials and Three-Body Choreographies* — Juan Manuel Sánchez Cerritos

Wednesday

- *From Equilibria to Collision: Complex Orbital Dynamics in a Simplified Asteroid Model* — Eva Tresaco
- *Solutions of the Three Body Problem from alignments* — Begoña Nicolás
- *Geometric properties of normally hyperbolic invariant manifolds and scattering maps for conformally symplectic systems* — Tere M-Seara
- *Asymptotic analysis of Hill-type orbits in the planar circular (2+2)-body problem* — Lennard Bakker
- *Fast diffusion pseudo-orbits in the R3BP and their shadowing* — Pablo Roldán

Thursday

- *Ejection-collision solutions in restricted N-body problems and KAM tori of near-rectilinear type* — Jesús Francisco Palacián
- *Brake orbits now and then* — Daniel Offin
- *Constructive KAM Theory for Quasiperiodic Lagrangian Tori: Algorithms and Applications to the Sitnikov Problem* — Pedro Porras Flores
- *Arnold diffusion in the full three-body problem* — Marian Gidea
- *The averaging method for Hamiltonian systems in the degenerate case* — Claudio Vidal Diaz

Friday

- *Jet transport techniques to study near-Earth object dynamics* — Luis Benet
- *Quasiperiodic Solutions of the Three-Body Problem: Misalignment in Exoplanetary Systems* — Patricia Yanguas
- *Study of black holes with a Higgs field at the horizon* — Víctor Castellanos Vargas

2 Workshop Schedule

Time	Mon	Tue	Wed	Thu	Fri
07:00–08:30	Breakfast	Breakfast	Breakfast	Breakfast	Breakfast
08:30–08:45	Registration	Breakfast	Breakfast	Breakfast	Breakfast
08:45–09:00	Presentation	Breakfast	Tresaco	Palacián	Breakfast
09:00–09:30	Moeckel	Stoica	Tresaco	Palacián	Benet
09:30–09:45	Moeckel	Stoica	Nicolás	Palacián	Benet
09:45–10:00	Moeckel	Stoica	Nicolás	Offin	Benet
10:00–10:15	Burgos	Jackman	Nicolás	Offin	Yanguas
10:15–10:30	Burgos	Jackman	Seara	Offin	Yanguas
10:30–10:45	Burgos	Jackman	Seara	Offin	Yanguas
10:45–11:00	Burgos	Jackman	Seara	Coffee	Yanguas
11:00–11:15	Coffee	Coffee	Seara	Porras	Coffee
11:15–11:30	Coffee	Coffee	Coffee	Porras	Coffee
11:30–11:45	Maderna	Zhu	Coffee	Porras	Castellanos
11:45–12:00	Maderna	Zhu	Bakker	Porras	Castellanos
12:00–12:30	Maderna	Zhu	Bakker	Gidea	Castellanos
12:30–12:45	Xie	Oliveira	Bakker	Gidea	Workshop Closing
12:45–13:00	Xie	Oliveira	Roldán	Gidea	
13:00–13:30	Xie	Oliveira	Roldán	Lunch	
13:30–13:45	Lunch	Lunch	Roldán	Lunch	
13:45–15:30	Lunch	Lunch / Photo	Free afternoon	Lunch	
15:30–16:30	Gonzaga	Escobar		Vidal	
16:30–17:30	Santoprete	Sánchez		Discussion Session	

The group photo will be taken on Tuesday from 3:20 PM to 3:30 PM in the CMO Conference Room.

3 Detailed program

Title: Scattering in the positive energy three-body problem

Speaker: Richard Moeckel

Abstract: In the three-body problem with positive energy, solutions which avoid triple collision have the property that the size of the triangle formed by the bodies tends to infinity as $t \to \pm\infty$. Furthermore, the triangles have well-defined asymptotic shapes $s_\pm$. The scattering problem asks which asymptotic shape s_+ can occur for a given choice of s_-. Previous work shows that this can be viewed as the problem of finding heteroclinic orbits connecting equilibrium points on a boundary manifold "at infinity", and some results were obtained for solutions which avoid collisions. The goal of this paper is to study the scattering effect of binary and near-triple collisions in a simple setting—the isosceles three-body problem. The details depend on the mass parameters, but in many cases a fixed isosceles initial shape s_- scatters to essentially all possible isosceles shapes s_+.

Title: When the space of quadratic forms meets Kepler's problem

Speaker: Jaime Burgos-García

Abstract: Several problems in Celestial Mechanics can be considered Keplerian problems, whether rotating or not, plus particular cases of homogeneous polynomials $Q(x, y, z)$ of degree two in three variables. These polynomials typically arise from different perturbations and have been studied for decades considering fixed values of their coefficients. In this talk we present a geometric approach that treats these systems as multi-parameter problems determined by the coefficients of $Q(x, y, z)$, and explores generic conditions for the existence of different invariant objects. We also show numerical explorations that visualize the evolution of Poincaré sections in the two-dimensional case. This is a collaborative work with A. Bengochea, J.A. Arredondo, and L. Franco.

Title: The natural decomposition of the Jacobi equation

Speaker: Ezequiel Maderna

Abstract: In the n-body problem, there are many subproblems. These arise whenever the gradient of the Newtonian potential leaves invariant some subspace of the configuration space (the gradient is taken with respect to the mass inner product). I will show that the Jacobi equation naturally decomposes over the orbits of such a subproblem. Examples of these subproblems include the case where all the bodies move in a fixed subspace, the case of isosceles triangles in the three-body problem, and the paradigmatic case of homographic solutions by central configurations (for the planar n-body problem). In this last case, we show that our decomposition is equivalent to the well-known Meyer–Schmidt decomposition of the linearization of the Euler–Lagrange flow. We present two applications: one to the linear stability of elliptic homographic motions, and the other to the hyperbolic scattering problem. Joint work with Renato Iturriaga.

Title: Degeneracy of four-body central configurations

Speaker: Zhifu Xie

Abstract: The degeneracy of central configurations in the planar N-body problem makes their enumeration problem difficult. To truly understand the bifurcations of central configurations, we work in the full configuration space, which also facilitates computer-aided methods. The degeneracy is always intertwined with the symmetry of the system of central configurations, which makes the problem subtle. By analyzing the Jacobian matrix of the system, we systematically explore a direct method to single out trivial zero eigenvalues associated with translational, rotational, and scaling symmetries, thereby isolating the non-trivial part of the Jacobian to study the degeneracy. The method is applied to explore the degeneracy of four-body central configurations. This is a joint work with Shanzhong Sun and Peng You.

Title: Bifurcations of Centered Symmetric Central Configurations

Speaker: Michelle Gonzaga dos Santos

Abstract: Finding new central configurations in the n-body problem and addressing long-standing finiteness questions remain significant challenges in celestial mechanics. In this context, the study of bifurcations has proven to be an effective approach, allowing the discovery of new central configurations from known degenerate ones. In many works, the masses of the bodies are treated as bifurcation parameters. In this talk, we extend the study of bifurcations within symmetric classes of central configurations involving four and five bodies. Specifically, we consider the centered equilateral triangle and the centered regular tetrahedron, with equal masses at the vertices and a body of arbitrary mass at the centroid. Previous results have traditionally focused either on a varying central mass under the Newtonian potential, or on varying vertex masses for homogeneous potentials with exponent strictly less than -1. In this work, we broaden this framework. For the centered equilateral triangle, considering homogeneous potentials with exponent less than $-1/5$, we classify all symmetric central configurations that arise as the central mass passes through its degenerate value. For the centered regular tetrahedron, we vary two or three vertex masses equally and classify all bifurcating configurations for exponents less than or equal to -1. In this case, we show that any configuration bifurcating from the degenerate one must remain symmetric when the varying masses are equal. Finally, we discuss bifurcations of the centered square configuration. Although this is a codimension-two problem, we show that barycentric coordinates provide an alternative and effective formulation of the associated system, opening promising new directions for discovering central configurations that bifurcate from this complex degenerate case.

Title: Conformal Relative Equilibria for Scaling Symmetries and Central Configurations

Speaker: Manuele Santoprete

Abstract: In this talk, we introduce the concept of conformal relative equilibria for vector fields that are conformally invariant under a scaling action. This framework extends the classical theory of relative equilibria to dynamics where scaling symmetries play a fundamental role. Focusing on Hamiltonian systems on exact symplectic manifolds, we define conformally symplectic actions and a conformal momentum map. Through an augmented Hamiltonian formulation, we derive a practical characterization of these equilibria and introduce a generalized notion of central configurations arising from scaling symmetries. While our results recover classical equations for the n-body problem, the theory is broadly applicable to systems with non-flat and conformally flat metrics, offering a new geometric perspective on scaling-invariant Hamiltonian dynamics.

Title: Symmetry and N-body dynamics deformation from the plane to the sphere

Speaker: Cristina Stoica

Abstract: We study the Newtonian n-body problem on a sphere of radius $R = 1/\epsilon$ as a smooth deformation of the planar problem, parametrised by the curvature $\epsilon > 0$ with $\epsilon = 0$ as the planar limit. Using Riemannian exponential coordinates at the North pole, we pull back the spherical Hamiltonian to a fixed phase space and show that it depends smoothly on ϵ, recovering the planar Newtonian Hamiltonian at $\epsilon = 0$. The symmetry algebra deforms simultaneously via the Inönü–Wigner contraction $\mathfrak{so}(3) \to \mathfrak{se}(2)$, carried out in a fixed basis that keeps the momentum maps and their flat limits explicit. In this framework we prove that every non-degenerate relative equilibrium and every non-degenerate relative periodic orbit of the planar problem continue smoothly to the sphere for small ϵ. Applications include Lagrange and regular n-gon relative equilibria, drifting configurations whose translational drift becomes rotational on the sphere, and the figure-eight choreography. We also discuss the persistence of hyperbolic invariant sets and their stable/unstable manifolds via structural stability.

Title: Collinear motions of three bodies in a three-sphere

Speaker: Connor Jackman

Abstract: For three point masses on a sphere subject to mutual attractions, we examine those motions for which the three bodies are at all times contained along some geodesic (great circle) of the sphere. Note that the geodesic containing the bodies is not required to be fixed. When the geodesic containing the three bodies sweeps out the three-sphere, we give a complete and explicit description of such motions. The corresponding question in the Euclidean case was solved by A. Wintner.

Title: On relative equilibria in the three-body problem on the two-dimensional hyperbolic space

Speaker: Shuqiang Zhu

Abstract: We study relative equilibria (RE) of the n-body problem on the two-dimensional hyperbolic space, focusing on the three-body case. We uncover a connection between the three classes of RE (elliptic, hyperbolic, and parabolic) and the three classes of vectors (time-like, space-like, and null) in Minkowski space. We introduce the inertia tensor for RE and explore its spectral properties. Our results include proofs of the non-existence of parabolic RE, fixed points, and RE under repulsive potentials. We show that all hyperbolic configurations are geodesic and, in a certain sense, constitute a subset of elliptic configurations. A central contribution is the development of criteria for three-body RE configurations in terms of mutual distances. Another contribution is the formulation of a counting problem for RE, together with its resolution in the geodesic case and in the three-body setting.

Title: The Veronese Geometry of Dziobek Configurations and Generic Finiteness for Homogeneous Potentials

Speaker: Thiago Dias Oliveira Silva

Abstract: The main goal of this talk is to present the proof of the generic finiteness of Dziobek central configurations for a homogeneous potential and the derivation of a uniform upper bound for their number. By exploiting the isomorphism between the Veronese variety and the determinantal variety associated with Dziobek configurations, we define the Dziobek-Veronese variety and apply the dimension of fibers theorem to analyze the projection from the space of configurations and masses to the space of masses. We prove that the fibers of this projection, representing the central configurations for a given mass vector, are finite for masses chosen outside a proper algebraic subvariety. Furthermore, we utilize that the Dziobek variety is defined by an intersection of quadrics to obtain a Bezout-type bound for the number of Dziobek configurations with fixed masses given by a power of 2 with exponent quadratic in n. Unlike previous estimates tailored to specific potentials, this bound depends solely on the dimension n. For instance, for the four-body problem, our bound reduces to 8192, which is lower than the bound of 8472 established by Moeckel and Hampton. This suggests that the complexity of counting Dziobek configurations for generic masses is governed primarily by the ambient geometry of the configuration space, rather than by the non-linearity of the interaction potential.

Title: Superintegrability and choreographic solutions in quadratic dihedral n-body Hamiltonians

Speaker: Adrián Escobar

Abstract: A solvable family of planar quadratic many-body Hamiltonian systems with dihedral symmetry is considered. The symmetry of the Hamiltonian separates the dynamics into normal-mode sectors, whereas choreographic motion requires an additional space–time symmetry: the bodies must follow a common curve with equal time shifts. This setting makes it possible to compare integrability, superintegrability, resonant closure, mode degeneracy, and choreographic motion analytically. Resonance leads to closed orbits, but choreography depends on a further compatibility between the active modes and the required time-shift symmetry. The mechanism is illustrated through four-, five-, and six-body examples.

Title: Homogeneous Potentials and Three-Body Choreographies

Speaker: Juan Manuel Sánchez Cerritos

Abstract: In this talk, I will present a variational approach to the construction of planar choreographies in the three-body problem with homogeneous potentials. The starting point is the minimization of the Lagrangian action on spaces of loops with prescribed symmetries, in the spirit of Chenciner–Montgomery and Marchal's collision-avoidance theorem. I will discuss how a suitable scale normalization leads to a homothety-invariant functional and provides collisionless periodic candidates within symmetric classes. Finally, I will explain how this framework relates to the existence of figure-eight-type orbits for homogeneous potentials.

Title: From Equilibria to Collision: Complex Orbital Dynamics in a Simplified Asteroid Model

Speaker: Eva Tresaco

Abstract: We consider a simplified asteroid model in which the gravitational potential is generated by a segment, and we analyse its dynamical consequences in comparison with standard polyhedral gravity models. We first validate the potential and show that it provides a computationally efficient approximation away from the singular set. We then study the resulting phase space structure, identifying equilibrium points and computing associated families of periodic orbits. Numerical continuation reveals a rich bifurcation structure, including heteroclinic and homoclinic connections between periodic solutions. Finally, we briefly discuss trajectories approaching the singular segment and outline how a local asymptotic reduction near the singularity could be used to better understand the behaviour of collision trajectories.

Title: Solutions of the Three Body Problem from alignments

Speaker: Begoña Nicolás

Abstract: In 1892 H. Poincaré explained a particular solution of the Three Body Problem in which the three bodies align periodically. These motions are typically known as syzygy solutions, and they cover different configurations as the bodies orbit one another. In this talk we describe this kind of solutions, how they can be numerically computed, and the insight they can provide in the study of some real systems. As an example, we analyse the case of the Kepler-16 system, a binary system of stars with a circumbinary exoplanet. This is a joint work with A. Jorba, M. Jorba-Cuscó and J. Gimeno.

Title: Geometric properties of normally hyperbolic invariant manifolds and scattering maps for conformally symplectic systems

Speaker: Tere M-Seara

Abstract: This talk is devoted to study conformally symplectic diffeomorphisms. Those arise in applications, such as mechanical systems with friction proportional to velocity.

We will focus on the existence of Normally hyperbolic invariant manifolds (NHIM) and their (un)stable manifolds, which are known to be important landmarks that organize long-term dynamical behaviour. We prove that a NHIM is symplectic if and only if the rates satisfy certain pairing rules and if and only if the rates and the conformal factor satisfy certain (natural) inequalities.

We also study scattering maps to the NHIM, which describe the homoclinic excursions to NHIMs. These maps give the trajectory asymptotic in the future as a function of the trajectory asymptotic in the past. We prove that the scattering maps are symplectic even if the dynamics is dissipative. We also show that if the symplectic form is exact, then the scattering maps are exact, even if the dynamics is not exact. We use these tools to detect instabilites in systems with small dissipation.

Title: Asymptotic analysis of Hill-type orbits in the planar circular (2+2)-body problem

Speaker: Lennard Bakker

Abstract: We study the symmetric continuation and asymptotic approximations of direct and retrograde Hill-type periodic solutions in terms of a mass ratio for a Hamiltonian system derived from the Planar Circular (2+2)-Body Problem. The unperturbed problem (mass ratio equal to zero) is the integrable Kepler problem in rotating coordinates plus a Coriolis term. We prove that circular periodic solutions of the unperturbed problem continue symmetrically to the perturbed problem (small positive mass ratio) while maintaining the energies of the unperturbed circular periodic solutions. We establish the elliptic stability of the symmetrically continued periodic solutions for all small mass ratios. We compute two-term asymptotic approximations of the symmetrically continued periodic solutions with prescribed energies using the Poincaré–Lindstedt method, where we determine a priori the dependence of the initial conditions and the frequencies of the symmetrically continued periodic solutions on the mass ratio. We illustrate how the initial conditions of symmetrically continued periodic solutions shift and how the graphs of these solutions deform from the unperturbed circular periodic solutions.

Title: Fast diffusion pseudo-orbits in the R3BP and their shadowing

Speaker: Pablo Roldán

Abstract: In a recent paper [DGR26], we considered the spatial R3BP around the equilibrium point L_1, and constructed diffusion pseudo-orbits with optimal diffusion time. The pseudo-orbits are obtained by iterating the so-called transition map on a suitable NHIM. They are chains of (finite-time) heteroclinic orbit segments, so they are not true orbits of the R3BP system. In this talk, we will explain how to construct true diffusion orbits that shadow the pseudo-orbits. We will use: the transition map; the Birkhoff normal form around L_1; and a (numerical) parallel shooting scheme. In contrast to common shadowing lemmas used in theoretical papers, we construct shadowing orbits of similar length than the pseudo-orbits. This is joint work with A. Delshams and M. Gidea. [DGR26]: Delshams, Gidea and Roldán: Semi-analytic construction of global transfers between quasi-periodic orbits in the spatial RTBP. Commun Nonlinear Sci Numer Simulat 152 (2026)

Title: Ejection-collision solutions in restricted N-body problems and KAM tori of near-rectilinear type

Speaker: Jesús Francisco Palacián

Abstract: We study ejection-collision (EC) solutions in several cases of the planar restricted N-body problem and in the spatial circular restricted three-body problem. These solutions are characterised by the fact that the particle with negligible mass ejects from one of the $N-1$ massive bodies and passes several times nearby before colliding with it. Working in the so-called lunar regime, where the infinitesimal particle moves near one of the massive bodies, the Hamiltonian function is a small perturbation of the Kepler problem. We apply a combination of regularisation, normalisation, symplectic reduction, and a KAM theorem for highly degenerate systems, proving the persistence of KAM 2-tori (3-tori for the spatial problem) filled with quasi-periodic motions of the small particle. Among these quasi-periodic motions we focus on the near-rectilinear ones. From these we identify those corresponding to EC solutions. For the planar problems, we always find four families of ejection-collision solutions, and for some configurations of the $N-1$ massive bodies this number increases, as shown in the examples. In the spatial case we find eight families of EC solutions corresponding to the vertical periodic and quasi-periodic solutions of the circular restricted three-body problem. All solutions are obtained analytically, providing the first terms of the corresponding expansions. This is a joint work with Aitor Ayape, Xabier Larequi and Patricia Yanguas.

Title: Brake orbits now and then

Speaker: Daniel Offin

Abstract: An historical account of brake orbits in classical Hamiltonian systems using global methods. Our approach uses the variational structure of the Jacobi metric and its geometric consequences. This involves an analysis of the co-geodesic flow as it intersects and reflects off the Hill's boundary. The continuation of this flow and its linearization provides a new approach to studying geometric properties such as stability.

Title: Constructive KAM Theory for Quasiperiodic Lagrangian Tori: Algorithms and Applications to the Sitnikov Problem

Speaker: Pedro Porras Flores

Abstract: We present a constructive a-posteriori KAM theorem for Lagrangian tori in quasiperiodically time-dependent Hamiltonian systems. The proof uses a quasi-Newton method that reduces the invariance error quadratically and naturally yields an efficient numerical algorithm. We apply the method to compute invariant tori in the Sitnikov problem, a classic restricted three-body configuration with elliptic primaries. This framework is suitable for rigorous computer-assisted proofs in celestial mechanics.

Title: Arnold diffusion in the full three-body problem

Speaker: Marian Gidea

Abstract: The full three-body problem, describing the motion of three celestial bodies under their mutual gravitational attraction, is one of the oldest unsolved problems in classical mechanics. The main difficulty comes from the presence of unstable and chaotic motions, which make long-term prediction impossible. In this work, we show that the full three-body problem exhibits a strong form of instability known as Arnold diffusion. We regard the planar full three-body problem as a perturbation of both the Kepler problem and the planar circular restricted three-body problem. We show that the system exhibits Arnold diffusion, in the sense that there is a transfer of energy—of an amount independent of the perturbation parameter—between the Kepler problem and the restricted three-body problem. Our argument is based on the topological method of correctly aligned windows, which is implemented into a computer-assisted proof. We demonstrate that the approach can be applied to physically relevant masses of the bodies, choosing a Neptune–Triton–asteroid system as an example. In this case, we obtain explicit estimates for the range of the perturbation parameter and for the diffusion time. Joint work with Maciej Capinski.

Title: The averaging method for Hamiltonian systems in the degenerate case

Speaker: Claudio Vidal Diaz

Abstract: In this talk, we present an extension of the Averaging Theorem for Hamiltonians to prove the existence of periodic solutions in Hamiltonian systems in the degenerate case. More precisely, we consider the family of analytic Hamiltonian functions $H_\varepsilon = H_0(I) + \varepsilon^\alpha H_1(I, y) + \varepsilon^{\alpha+p} H_2(I, y) + \cdots$, with $\alpha \in \{1, 2\}$ and $p \in \mathbb{N}$, where ε is a small parameter; the conjugate variable to the action I_c is θ, which is an angular variable. The point y^* is a degenerate (isolated) critical point of H_1 at the fixed level h, i.e., $H_\varepsilon = h$, $\nabla H_1(y^*) = 0$ and $\det(\mathrm{Hess}\, H_1(y^*)) = 0$. If the kernel of $\mathrm{Hess}\, H_1(y^*)$ (with respect to the variable y) has dimension k, where $1 \le k \le 2(n-2)$, then we prove that from the point y^* there can bifurcate at most $2k$ different periodic solutions of the full system. Furthermore, we provide an approximation of the characteristic multipliers associated with these periodic solutions. We present an illustrative example of our main results, showing that the maximal number of periodic solutions, given by $2k$, can indeed be attained.

Title: Jet transport techniques to study near-Earth object dynamics

Speaker: Luis Benet

Abstract: In this talk, I will describe how jet transport techniques can be applied to the study of the dynamics of near-Earth objects (NEOs). Jet transport exploits high-order polynomial expansions around nominal parameters based on automatic differentiation techniques. I will describe its application to the preliminary orbit determination problem, which consists precisely in finding nominal orbits based on observations including associated uncertainties, and in estimating the impact probability (e.g. with the Earth) of a recently discovered NEO. This work is done in collaboration with L. E. Ramírez-Montoya and J. A. Pérez-Hernández.

Title: Quasiperiodic Solutions of the Three-Body Problem: Misalignment in Exoplanetary Systems

Speaker: Patricia Yanguas

Abstract: The ongoing discovery of exoplanetary systems has significantly broadened our understanding of planetary orbital configurations around stars. The Herculis 14 system consists of a central star accompanied by two observed planets that exhibit a mutual orbital misalignment of approximately 35 degrees. In this work, we employ the Hamiltonian formulation of the spatial three-body problem to compute quasiperiodic solutions that are consistent with the dynamical structure of the Herculis 14 system.

Title: Study of black holes with a Higgs field at the horizon

Speaker: Víctor Castellanos Vargas

Abstract: In the study of black holes with a Higgs field, a Lotka–Volterra model appears naturally as a differential system in the space. In this talk, we provide a qualitative analysis of the flow of this system, describing the α-limit set and the ω-limit set of all orbits.
